



Introduction to Cybersecurity

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Lecture 5

Lecture Rules



Cryptography Basics

Part II

Lecture Outline

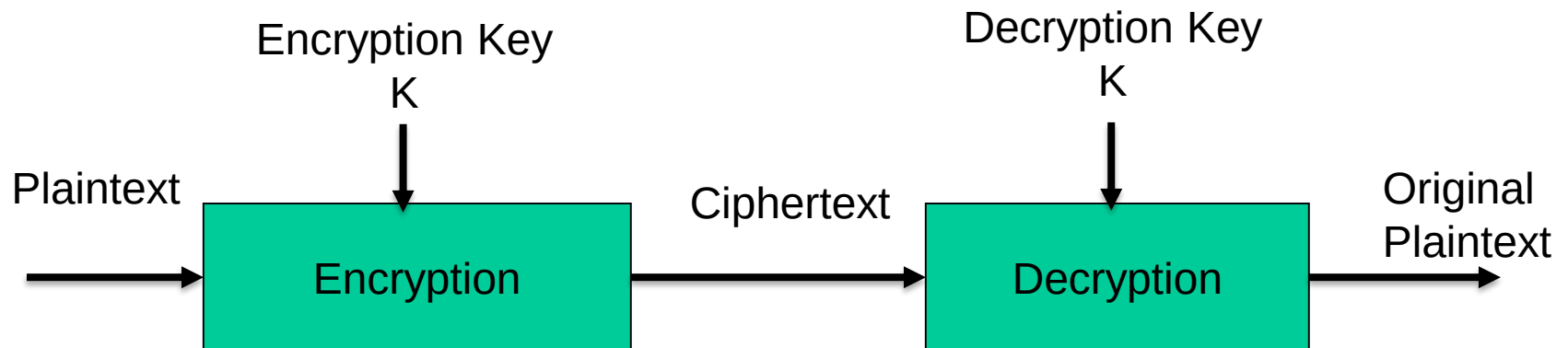
- Overview
- Symmetric Algorithms
- Asymmetric Algorithms (Public-Key Algorithms)
- Cryptanalysis
- Cryptanalysis Approaches:
 - Brute force Attacks
 - Cryptanalytic Attacks
- Caesar Cipher

Overview

- The fundamental objective of **cryptography** is to enable two people to **communicate** over **unsecured** channel in such a way that third party cannot **understand** what is being said.
- **This channel could be a telephone line or computer network (internet).**

Symmetric Algorithms

- **Algorithms** in which the **key** for **encryption** and **decryption** are the **same**.
- **Notations**
 - Message, **M**
 - Ciphertext, **C**
 - Key, **K**
 - Encryption function, $E_k(M) = C$
 - Decryption function, $D_k(C) = M$
- **Identity:** $D_k(E_k(M)) = M$



Symmetric Algorithm Types

1. Block Ciphers

- Encrypt data **one block** at a time (typically **64 bits**, or **128 bits**).
- Used for a **single message**.

2. Stream Ciphers

- Encrypt data **one bit** or **one byte** at a time.
- Used if data is a **constant stream** of **information**.

Symmetric Encryption – Key Strength

- Strength of algorithm is determined by the **size of encryption key**
 - A longer encryption key generally makes **ciphertext** more **difficult** to **crack**.
- Key length is expressed in **bits**
 - Typical key sizes vary between **48 bits** and **448 bits**.
- Set of possible keys for a cipher is called **key space**
 - For **40-bit** key there are **2^{40}** possible keys.
 - For **128-bit** key there are **2^{128}** possible keys.
 - Each additional bit added to the key length **doubles** the **security**.

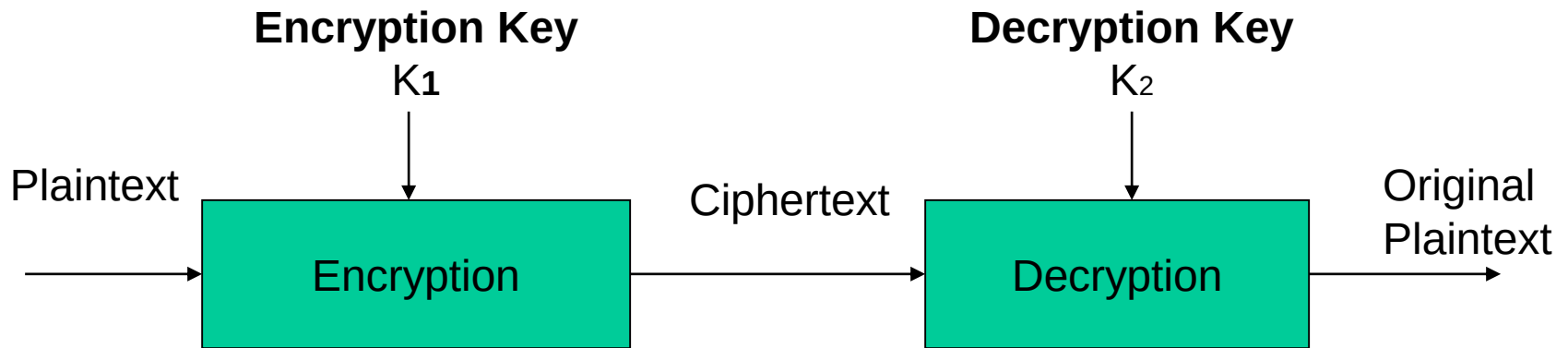
Symmetric Algorithms

- **DES (Data Encryption Standard)**
 - **Block size** is **64 bits**
 - **Key size** is **56 bits**
- **3DES (Triple Data Encryption Standard)**
 - Using DES **three times** on a **given message**
 - 3DES is a popular **symmetric block algorithm**.
- **AES (Advanced Encryption Standard)**
 - Block size is 128 bits.
 - **Key** sizes of **128**, **192**, and **256** bits.

Asymmetric Algorithms

Public-Key Algorithms

- Algorithms in which **different keys** are used for **Encryption** and **Decryption**.
- **Encryption key** (public key), K_1
- **Decryption key** (private key), K_2
- Encryption function, $E_{k1}(M) = C$
- Decryption function $D_{k2}(C) = M$



- For **digital signatures**, messages are **encrypted** with the **private key** and **decrypted** with the **public key**.

Asymmetric Algorithms

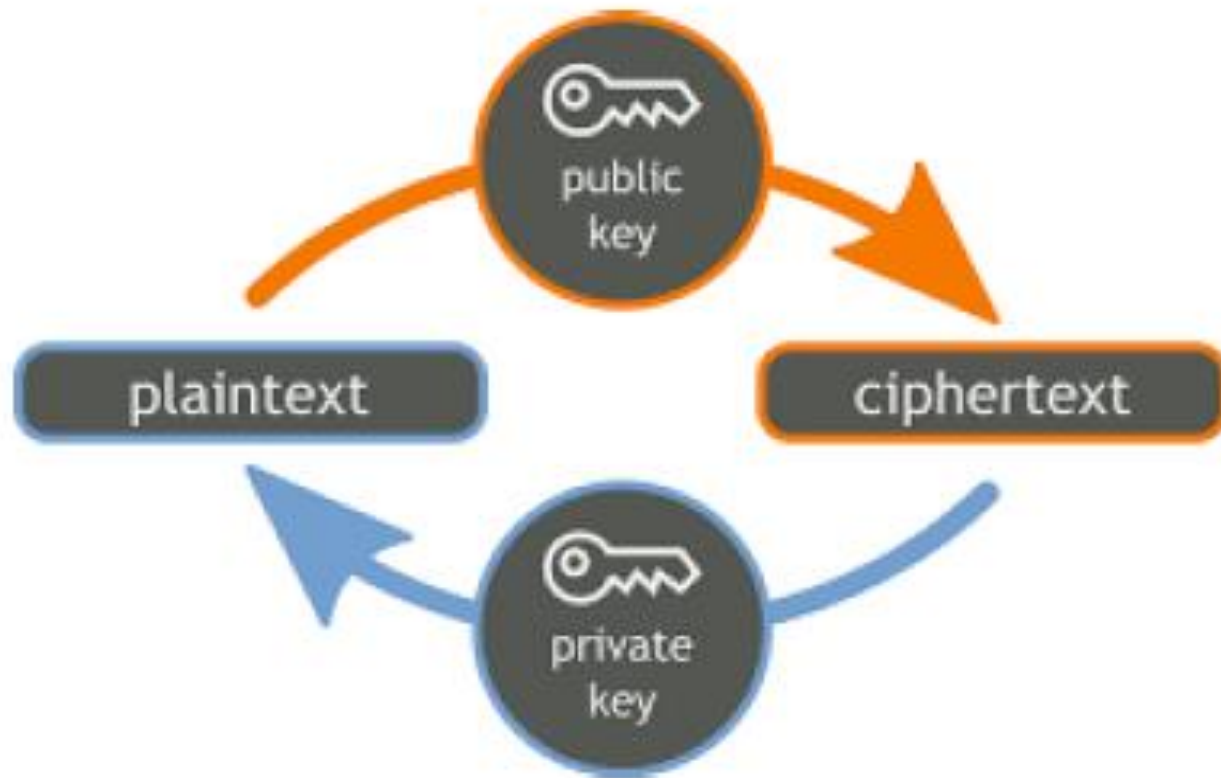
Public-Key Algorithms

- **Decryption key cannot be calculated** from **encryption key** in any **reasonable amount of time**.
- **Private Key:** Kept secret by the owner.
- **Public Key:** Shared with everyone.

Asymmetric Algorithms

Public-Key Algorithms

- Examples: RSA(Rivest, Shamir and Adleman), DSA(Digital Signature Algorithm), Diffie-Hellman



Cryptanalysis

- The art and science of breaking cipher text is called cryptanalysis.
- Cryptanalysis deals with finding the encryption key for breaking cryptographic algorithms without the knowledge of the encryption
- Cryptanalyst: a person who breaks cryptographic codes . Also referred to as “the attacker”

Cryptanalysis

- Science of recovering plaintext of a message without access to the key.
- An **attempted** cryptanalysis is called an **attack** where cipher **(algorithm)** and **implementation** are known.
- The main **objective** is to **recover key** not **just message**.

Complexity Measures of An Attack

- **Data complexity**
 - **Amount** of **data** needed as input to the **attack**
- **Processing complexity** (also called **work factor**)
 - **Time needed** to perform the **attack**
- **Storage requirements**
 - **Amount** of **memory** needed to do the **attack**
- So, the **complexity** of an attack is taken to be the **minimum** of the **three** factors.

Cryptanalysis Approaches

1. Brute-force attacks

- Try every **possible** key on a **piece** of **ciphertext** until an **intelligible** translation into **plaintext** is **obtained**.
- **On average, half** of all possible **keys** must be tried to **achieve success**.

2. Cryptanalytic attacks

- Try on the **nature** of the **algorithm** **plus** some **knowledge** of **characteristics** of the **plaintext** or even some **sample plaintext-ciphertext pairs**.

Brute Force Attacks

- Possible to **try every key to know / recognize plaintext.**

Key Size (bits)	Number of Alternative Keys	Time required at 1 decryption/ μ s	Time required at 10^6 decryptions/ μ s
32	$2^{32} = 4.3 \times 10^9$	$2^{31} \mu\text{s} = 35.8 \text{ minutes}$	2.15 milliseconds
56	$2^{56} = 7.2 \times 10^{16}$	$2^{55} \mu\text{s} = 1142 \text{ years}$	10.01 hours
128	$2^{128} = 3.4 \times 10^{38}$	$2^{127} \mu\text{s} = 5.4 \times 10^{24} \text{ years}$	$5.4 \times 10^{18} \text{ years}$
168	$2^{168} = 3.7 \times 10^{50}$	$2^{167} \mu\text{s} = 5.9 \times 10^{36} \text{ years}$	$5.9 \times 10^{30} \text{ years}$
26 characters (permutation)	$26! = 4 \times 10^{26}$	$2 \times 10^{26} \mu\text{s} = 6.4 \times 10^{12} \text{ years}$	$6.4 \times 10^6 \text{ years}$

Cryptanalytic Attacks

1. Ciphertext only

- only know **algorithm** & **ciphertext**, is **statistical**, know or can **identify plaintext**.

2. Known plaintext

- **know/suspect plaintext** & **ciphertext**.

3. Chosen plaintext

- select **plaintext** and obtain **ciphertext**.

4. Chosen ciphertext

- select **ciphertext** and obtain **plaintext**.

5. Chosen text

- select **plaintext** or **ciphertext** to **encrypt/decrypt**.

Caesar Cipher

- A type of **mono-alphabetic substitution cipher** where each letter of the plaintext is shifted a **fixed number** of places down the alphabet.
- This type of **substitution Cipher** is named after **Julius Caesar**, who used it to communicate with his generals.

Caesar Cipher Cont.

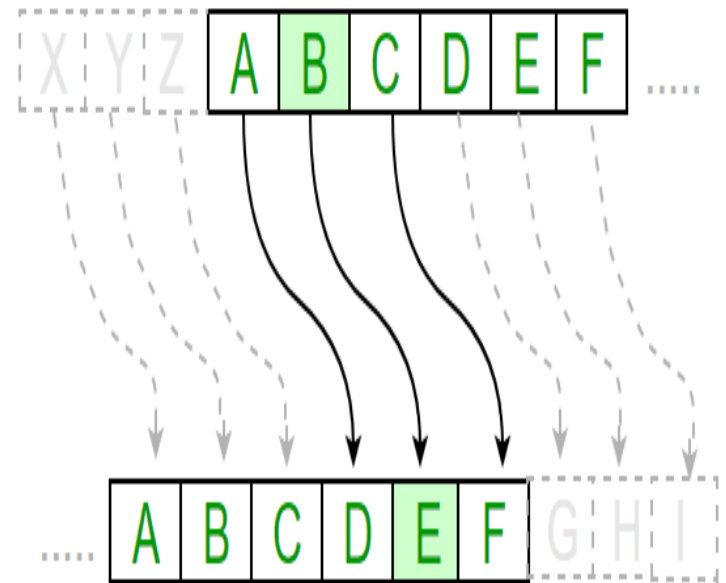
- The encryption can be represented using modular arithmetic by first transforming the letters into numbers, according to the scheme, $A = 0, B = 1, \dots, Z = 25$.
- Encryption of a letter by a **shift n** can be described mathematically as follows:

Encryption Phase with shift n :

$$E_n = (x + n) \bmod 26$$

Decryption Phase with shift n :

$$D_n = (x - n) \bmod 26$$



Caesar Cipher Encryption

- This type of cipher is a form of **symmetric encryption** as the **same** key can be used to both **encrypt** and **decrypt** a message.
- **Ex:**
 - Let's say our message is “**CAT**” using **Caesar Cipher encryption** with a **key of 5**, (shift each letter **forward 5 positions(+5)**):
 - C = H
 - A = F
 - T = Y
 - **So "CAT" becomes "HFY".**

Caesar Cipher Decryption

- **Decryption** of ciphertext with known Key

- Inverse of encryption

- **Ex:**

To decrypt "HFY" using **Caesar Cipher decryption**, we use a **key of 5** (shift each letter **backwards 5 positions (-5)**) :

- H = C

- F = A

- Y = T

- **So "HFY" becomes "CAT".**

Caesar Cipher Cont.

- **Cryptanalysis**

- **Exhaustive** key search (brute-force attack) is available.
 - **The key space is too small:** Only have **26 possible keys**, where A maps to A,B,..., Z.
- A **brute force search**, can **try each** of the keys (**shifts**) in turn, until you get the **original message (Plaintext)**.
- Given **ciphertext**, just try **all shifts** of letters.
- **Ex:**
- "GCUA VQ DTGCM" when **cryptanalysis**, it gives **"EASY TO BREAK"**, with a shift of **-2**

Example

- For getting a new secret key table, where $k=3$ to left. So the table will be:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

- This cipher shift alphabet three letters to the left.

Task (10 Min)

1. Assume $A=0, B=1, \dots, Z=25$, with $K=16$ to **encrypt** the following **plaintext** using **Caesar Cipher**:

- **Send Secret message**

2. Assume $A=, 0, B=1, \dots, Z=25$, with $K=11$ to **decrypt** the following **ciphertext** using **Caesar Cipher**:

- **hp htww xppe**

thanks for
listening!